

Earlier Forms of 3D Technology

In the mid 19th century, Charles Wheatstone discovered that viewing a pair of similar images side-by-side can give the impression of three-dimensionality. The images are taken by two cameras that are slightly separated. This way, the photographs mimic what each one of our two eyes would see in reality.

In the late 19th and early 20th century, the stereoscope was invented to address these issues. The stereoscope used lenses that merged the two images into one, giving the effect of a 3D image without straining the eyes. These glasses with red and blue lenses, when used with special photographs called anaglyph images, created the illusion of depth.

Traditionally, anaglyph images were taken using two cameras, one with a red filter and one with a blue filter. These images were then combined to form a single picture or anaglyph image. When viewed without 3D glasses, these images looked blurry and discoloured. These red and blue lenses tricked the brain into seeing a 3D image.

a fantastic 3D feel," shares Rajkumar. The camera also allows users to edit 3D images—a feature that is useful for landscape photography or, conversely, macro shots, where the 3D effect can be very strong.

Webcam. Taking a side step from cameras to webcams, Minoru is the world's first 3D webcam, allowing users to appear in 3D over Skype or any instant messaging service, as long as the person watching the user is wearing 3D glasses (five pairs come free with the webcam). Minoru also allows to shoot 3D movies and pictures and upload them on YouTube; the site has recently made way for 3D tech.

Television. Samsung has added the 3D technology to its high definition (HD) digital light projection (DLP) TV. It re-engineered its HD DLP TV to deliver 30fps video to each eye (requires stereoscopic 3D glasses in order to work). For this, Samsung increased the refresh rate of its HD DLP TV to 120 Hz. Now Samsung plans to apply the technology to its LCD and plasma displays as well.

Panasonic has also debuted the 3D full HD (3DFHD) plasma home theatre system—which the company claims to be the world's first—at CEATEC Japan 2008 in October 2008, followed by the US debut at CES 2009. LG had also showcased two 3D TVs at CES 2009—140cm and 152cm models—each loaded with HDMI and LG's top image processing tech.

3D screens are not just confined to

TVs. Consumers can also hope to see 3D products from Sanyo, Sony, Fujitsu and Mitsubishi, who promise to launch 3D-enabled products soon, though all won't be wall-size TVs. There could be small 3D screens that fit on cellphones or portable media players that don't require 3D glasses.

3D printer. DesignTech Systems Ltd has launched uPrint, a personal 3D printer from its Dimension series, in India. It provides users a networked, desktop-sized 3D printer, capable of building durable and accurate models ready for testing. Designed for the desktop, uPrint requires only a 63.5x66cm footprint.

3D films. Anaglyph images are widely adopted by the Hollywood film industry.

3D movies from Walt Disney, Sony Pictures and Dreamworks have been great hits at the box office. Other movie studios are also preparing 3D versions of their forthcoming and past movie releases. Although viewers still have to wear 3D glasses but these are stereoscopic glasses, not the old anaglyphs with red and green plastic lenses on each eye.

What's the future for 3D?

"We have already seen 3D applications in motion pictures, outdoor media, television, virtual worlds (video games), etc. But one wonders in which direction the 3D technology is headed?" asks Rajkumar. "Considering the technological development aspect, we can foresee a lot of

convergence happening in this area. We can anticipate a virtual world of 3D images, which could be the next big thing. This would open up various possibilities, including advanced research in space related matters," he explains. However, this is just a peep into the future and is all in the realm of conjecture, he affirms.

"Undoubtedly, this new imaging system looks exciting that may interest many a technologist into bundling the 3D functionality within their own innovations to add a competitive new dimension to their products," says Ghosh.

Consumers may also expect more emphasis on 3D gaming to come through next year or so—big names have shown interest in the technology, including the likes of Sony. The S3D Gaming Alliance has been formed to help developers with their 3D gaming projects. Blitz Games Studios' 'Invincible Tiger: The Legend of Han Tao' is arguably one of the first games to support modern 3D technologies. We can expect many more to follow.

Hitachi released a 3D mobile phone in Japan at the beginning of the year. Acer has come up with a 3D laptop. With the help of 3D technology, surgeons can now see inside the human body, without making incisions. 3D airspace simulators enable airtraffic controllers to view the activity of an airplane in 3D. The technology also gives scientists the ability to see viruses in 3D. So, we can expect the 3D technology to move into newer areas in the not too distant future.

Is it worth investing in?

Interest of consumers in the technology certainly seems to have gone up but, at the moment, 3D kits are quite expensive. Until prices come down, the technology would largely be used in cinemas rather than homes. But as more and more manufacturers get involved to find new application areas, competition will bring the prices down. It seems, the future belongs to 3D technology! ●